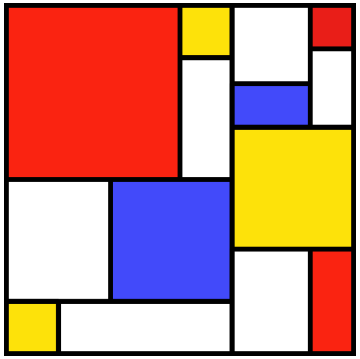
SLiM

Workshop Series



Callbacks II (`mateChoice()`)

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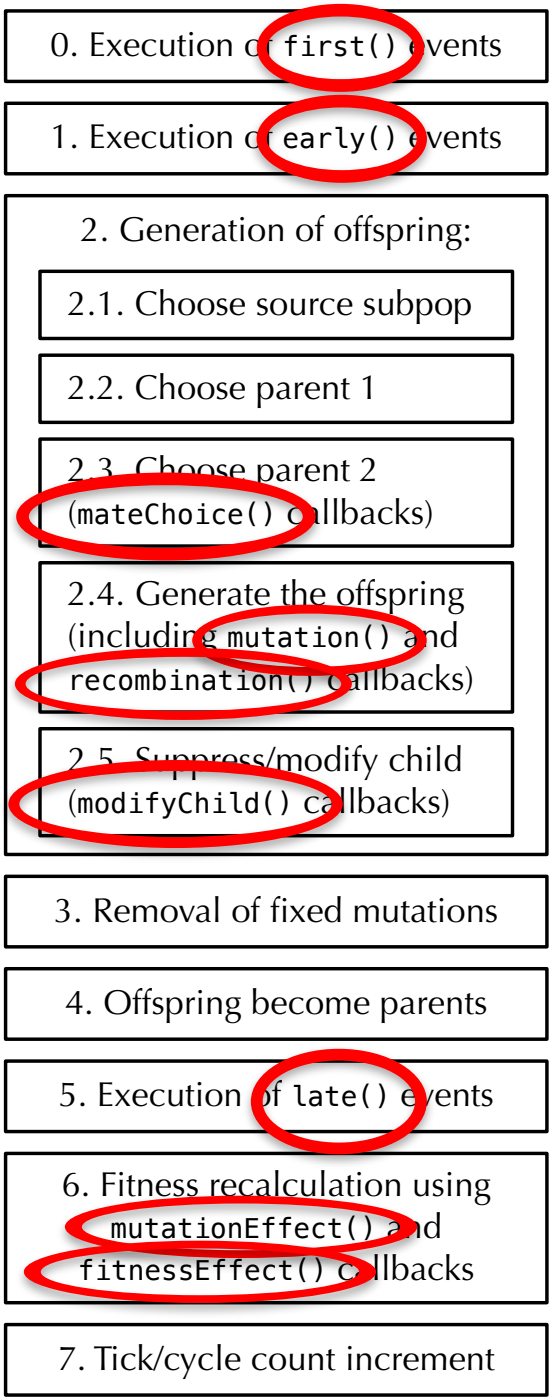
The WF model's tick cycle



The WF Tick Cycle

- Callbacks:
  - `first()` and `early()` events
  - `mateChoice()` callbacks
  - `mutation()` callbacks
  - `recombination()` callbacks
  - `modifyChild()` callbacks
  - `late()` events
  - `mutationEffect()` callbacks
  - `fitnessEffect()` callbacks
  - `interaction()`, `survival()`, `reproduction()`, ...

The WF model's tick cycle

0. Execution of `first()` events

1. Execution of `early()` events

2. Generation of offspring:

   2.1. Choose source subpop

   2.2. Choose parent 1

   2.3. Choose parent 2 (`mateChoice()` callbacks)

   2.4. Generate the offspring (including `mutation()` and `recombination()` callbacks)

   2.5. Suppress/modify child (`modifyChild()` callbacks)

3. Removal of fixed mutations

4. Offspring become parents

5. Execution of `late()` events

6. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks

7. Tick/cycle count increment

`mateChoice()` callbacks

- called during mate choice
- override default mating
- often used to implement:
  - assortative mate choice
  - sequential mate choice
  - monogamy
  - spatial mate choice
  - parental infertility

The WF model's tick cycle

0. Execution of `first()` events
1. Execution of `early()` events
2. Generation of offspring:
   - 2.1. Choose source subpop
   - 2.2. Choose parent 1
   - 2.3. Choose parent 2 (`mateChoice()` callbacks)
   - 2.4. Generate the offspring (including `mutation()` and `recombination()` callbacks)
   - 2.5. Suppress/modify child (`modifyChild()` callbacks)
3. Removal of fixed mutations
4. Offspring become parents
5. Execution of `late()` events
6. Fitness recalculation using `mutationEffect()` and `fitnessEffect()` callbacks
7. Tick/cycle count increment

`mateChoice()` callbacks

```
mateChoice([<subpopID>])
{
    ...
}
```

- apply to *one or all* subpops
- called for each **focal parent**
- return value:
  - a vector of mating weights
  - a singleton `Individual`
  - `float(0)` (no suitable mate)
  - `NULL` (use default weights)

The WF model's tick cycle

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5. Execution of `late()` events

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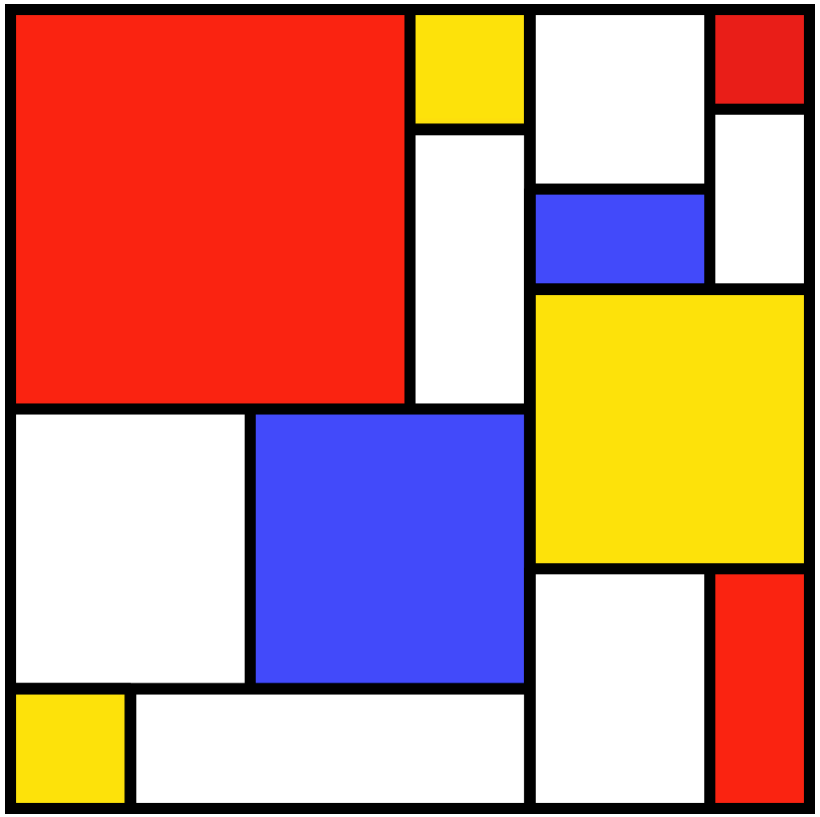
`mateChoice()` callbacks

- there is a *focal first parent*
- there are also default mating weights, based on fitness
- **pseudo-parameters**:
  - `individual`
  - `subpop`
  - `sourceSubpop`
  - `weights`



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